CRYPTOCURRENCY INVESTMENT FORECASTING

Presented by:
Aryan Garg (21MA10013)

OVERVIEW

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INTRODUCTION

- Cryptocurrencies are decentralized digital assets operating on blockchain technology, enabling secure and transparent transactions.
- Bitcoin and Ethereum are among the most prominent cryptocurrencies, representing a significant share of the market capitalization.

Project Vision:

- Empowering smarter investment decisions in cryptocurrencies through accurate and innovative forecasting models.

Project Mission:

- To leverage machine learning techniques for forecasting cryptocurrency prices with precision and reliability.
- To provide insights that guide investors in selecting the cryptocurrency for maximizing returns.

MOTIVATION

- **High Volatility:** Cryptocurrencies are highly volatile, making it essential to predict price movements for informed investment decisions.
- **Investment Opportunities:** Accurate forecasts help investors assess potential risks and returns, supporting portfolio diversification strategies.
- **Practical Applications:** Enhances algorithmic trading, financial planning, and risk management for individual and institutional investors.
- **Advancing Research in Volatile Markets:** This study highlights the application of advanced forecasting techniques in dynamic markets and opens avenues for exploring additional cryptocurrencies and hybrid models in future work.

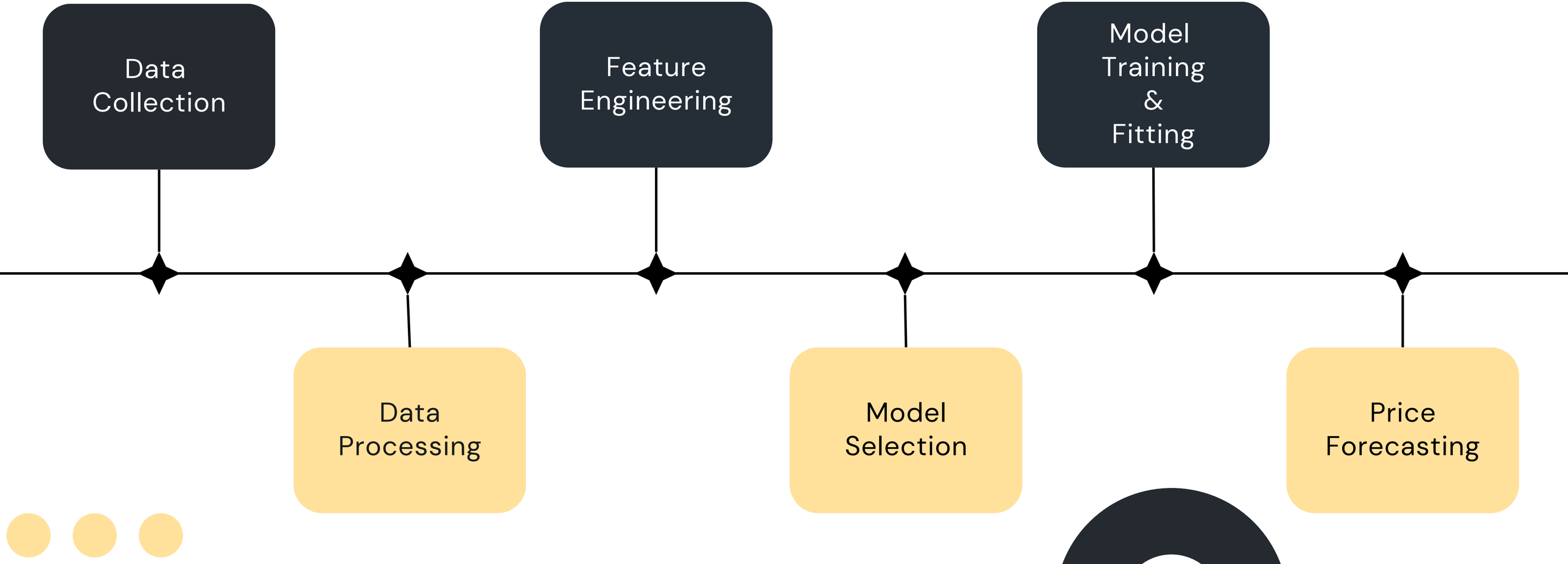


IDEATION PROCESS

- **Researching Cryptocurrencies:** Conducted an in-depth study of Bitcoin and Ethereum to understand their financial significance, volatility, and historical trend
- **Exploring Forecasting Models:** Delved into academic research and industry practices to identify potential forecasting models suitable for time series data
- **Identifying the best-fit Model:** Conducted a comparative analysis of ML models to determine the most accurate model based on RMSE and prediction reliability.
- **Predicting Future Prices:** Applied the best-fit model to forecast future prices for Bitcoin and Ethereum.
- **Investment Decision:** Compared forecasted returns for Bitcoin and Ethereum to identify the cryptocurrency with maximum potential returns.

METHODOLOGY

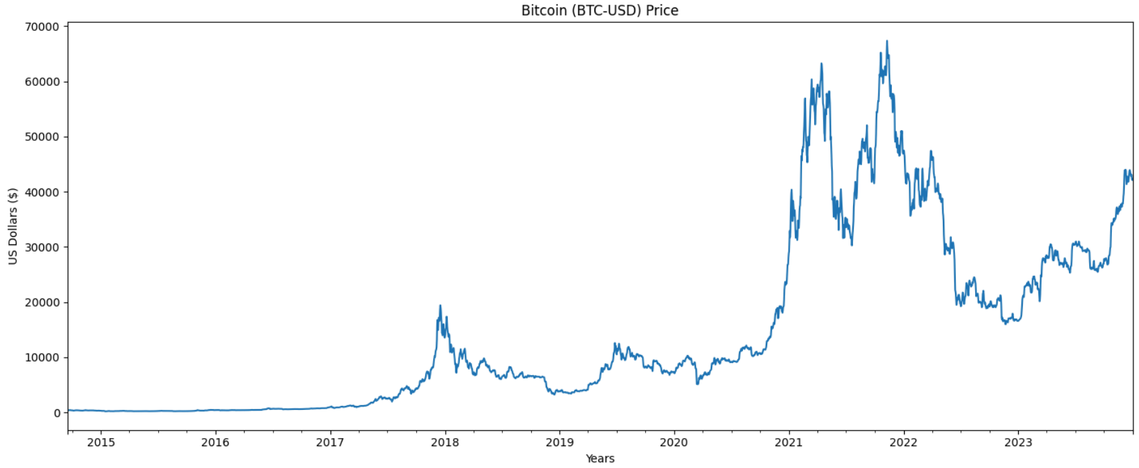
Approach to time series forecasting is through advanced machine learning and statistical models such as ARIMA, Rolling Forecast, ARIMA, LSTM, and XGBoost.

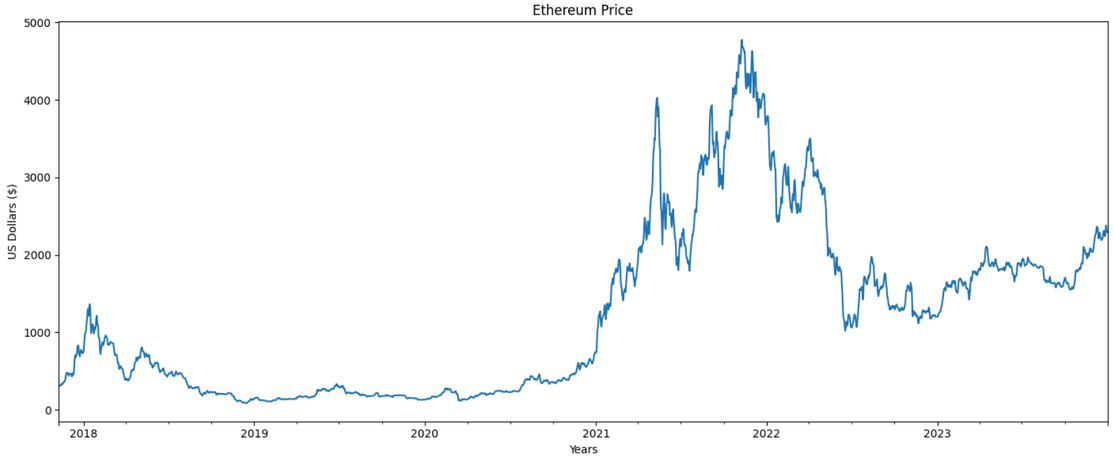


DATA COLLECTION & PROCESSING

- The dataset used for the project is sourced from Yahoo Finance
- It contains daily historical price data for Bitcoin and Ethereum.
- Calculated “weighted_price” using features features such as Open, High, Low, Close.

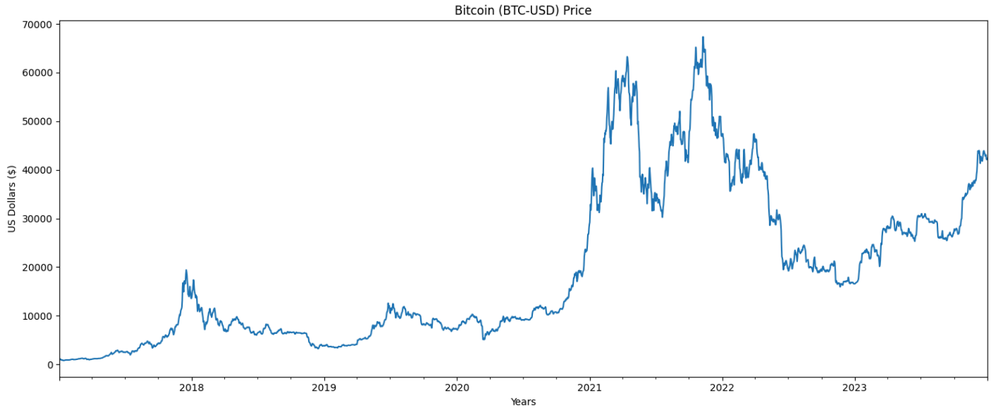
$$Weighted\ Price = (High + Low + Open + Close)/4$$

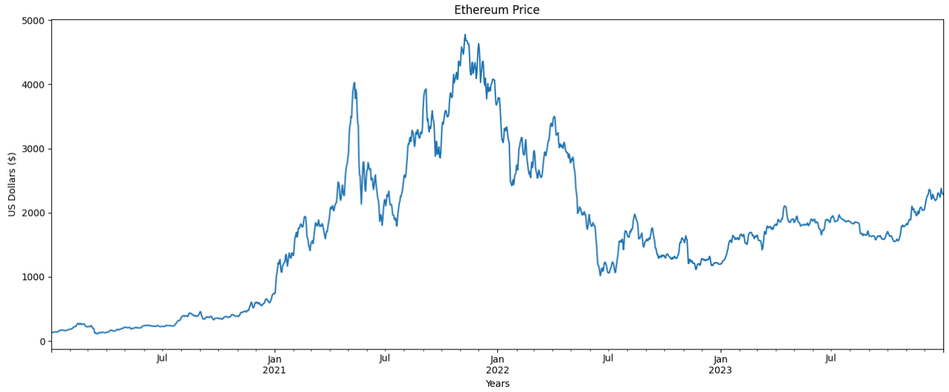




FEATURE ENGINEERING

Feature Engineering refers to the process of creating, modifying, or selecting features (input variables) and also selecting the precise amount of dataset that enhance the performance





Features like "Day of the Week", "Month", "Year", etc. needed to be extracted from the “Date” field while using the XGBoost Model and generating new features from the raw data make the data more informative.

MODEL SELECTION & TRAINING

The following models were used:

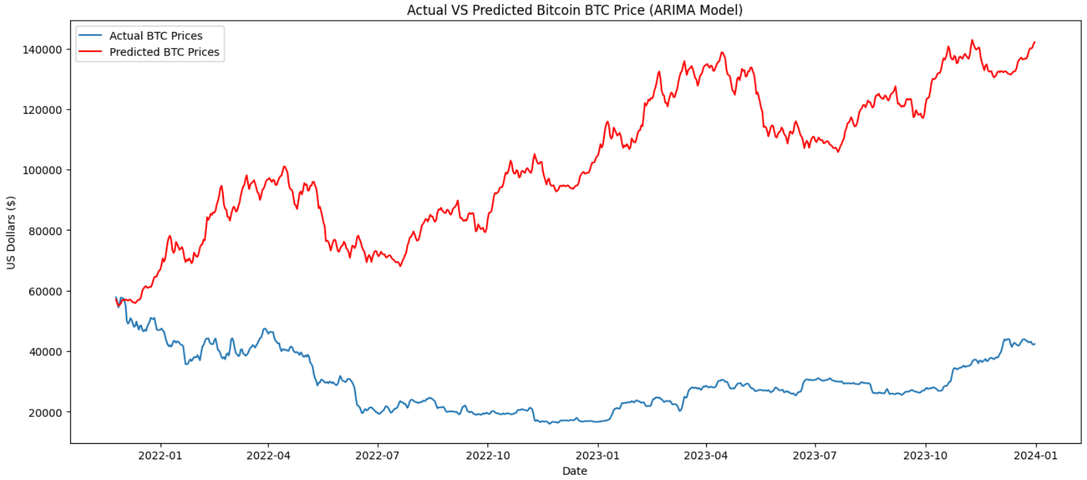
- ARIMA: The model parameters (p, d, q) were determined using the ACF and PACF plots.
- Rolling Forecast ARIMA: An extension of ARIMA that updates the training data dynamically allowing predictions to adapt to recent trends.
- LSTM: The dataset needed to be reshaped as stated before and use of 3 layers leads to increase in time but the model is properly trained. Activations used to build layers was researched.
- XGBoost: It was used with the objective of minimizing the squared erro (reg:squarederror) and keeping the value of estimators high (1000) to improve the performance

Training Dataset = 70%

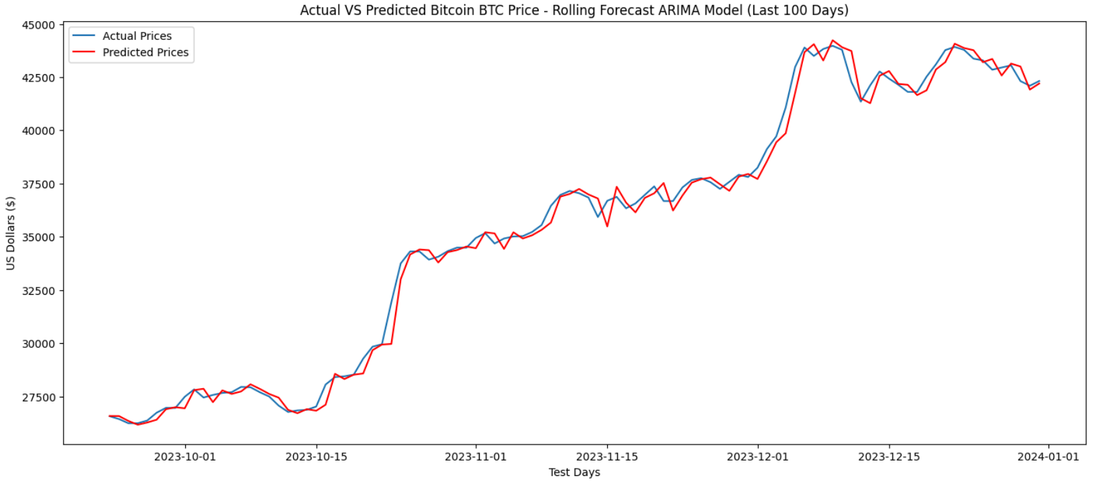
Testing Dataset = 30%

FORECASTS-BTC

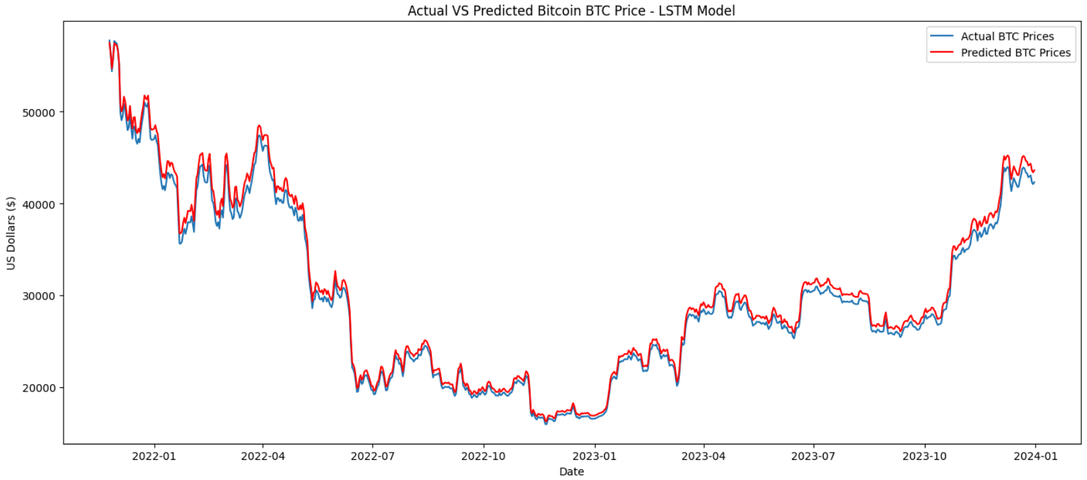




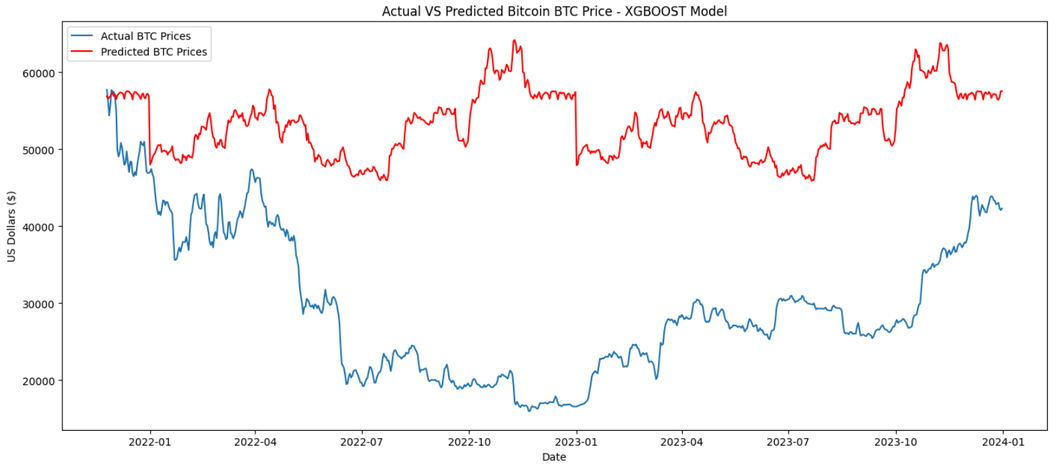
ARIMA



Rolling Forecast ARIMA



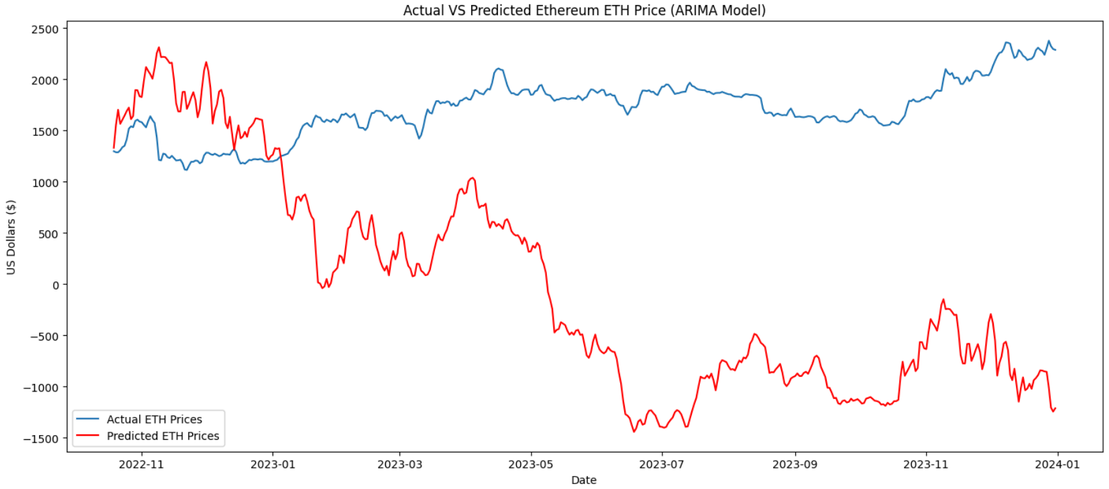
LSTM



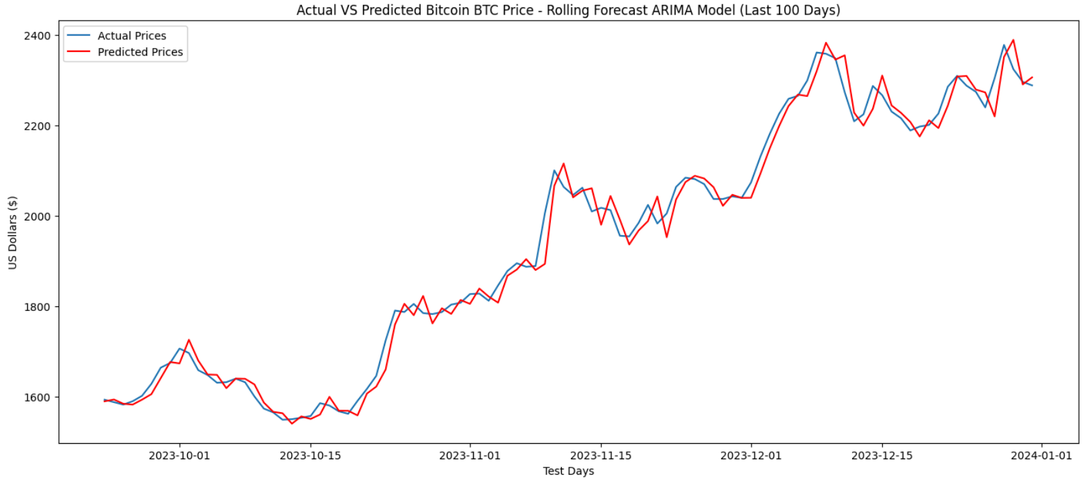
XGBoost

FORECASTS-ETH

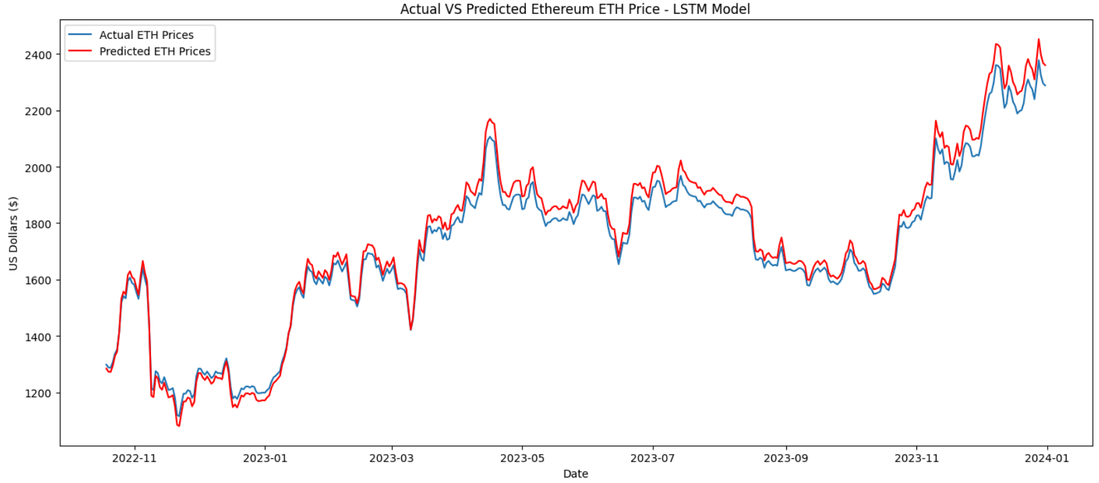




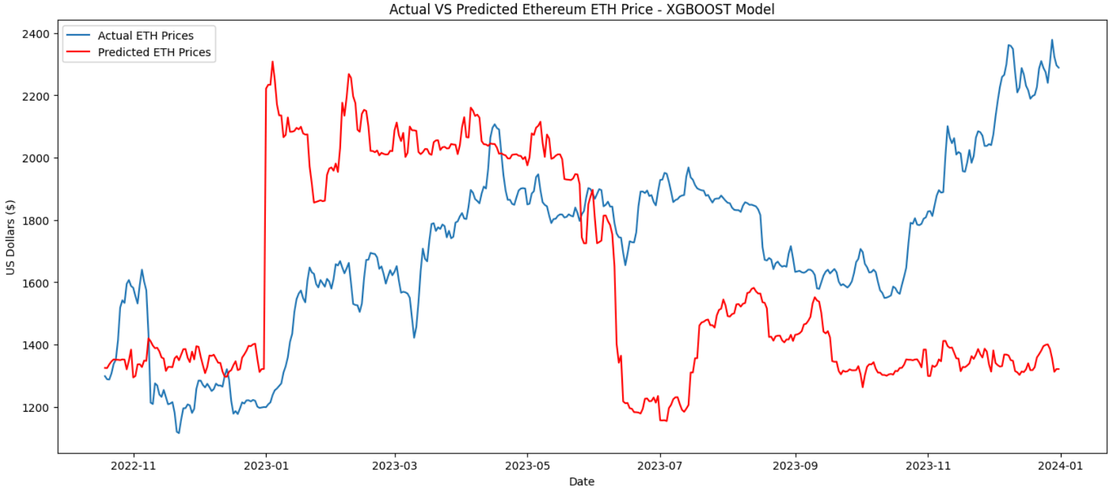
ARIMA



Rolling Forecast ARIMA



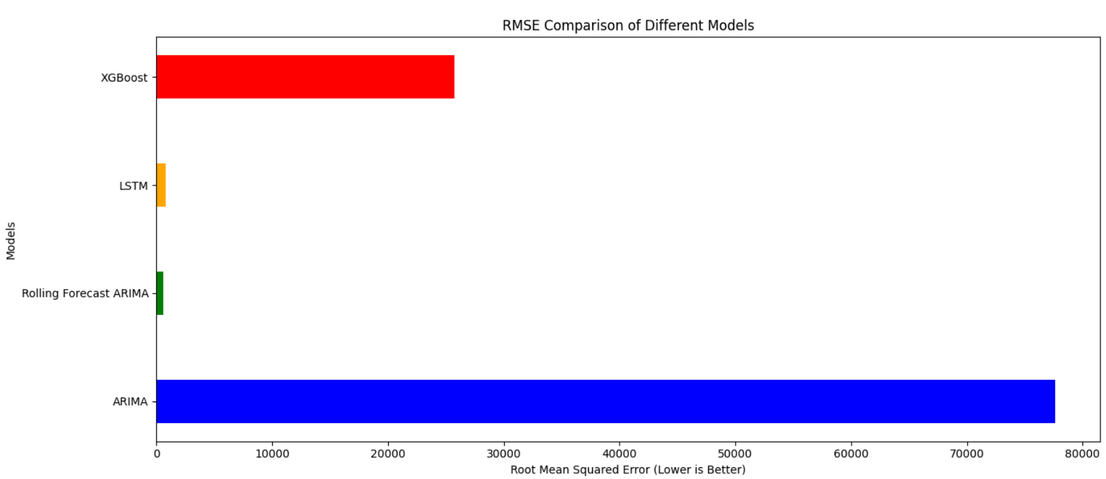
LSTM



XGBoost

BEST FIT MODEL

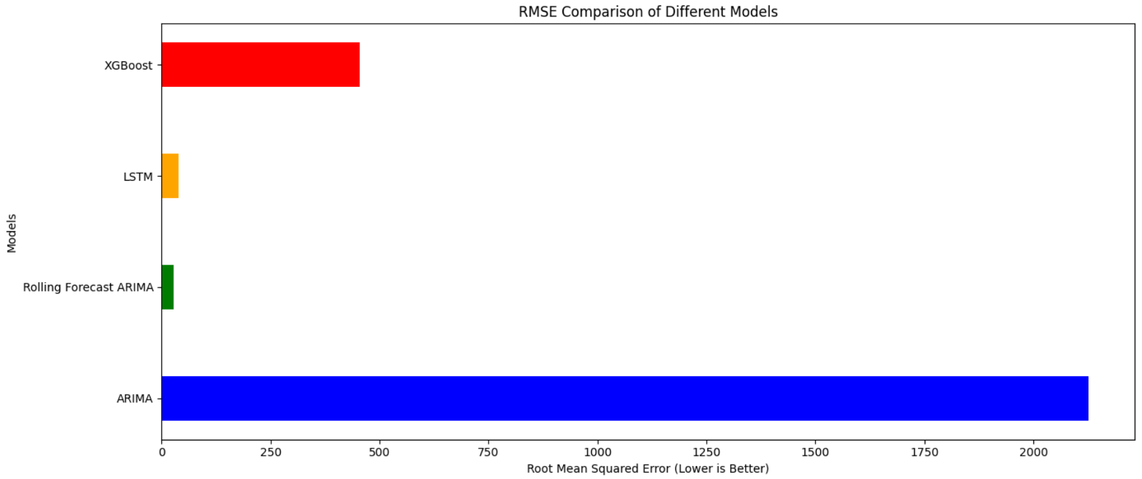
BITCOIN



LSTM Model is the best-fit model

Test RMSE for LSTM on BTC: **829.669**

ETHEREUM

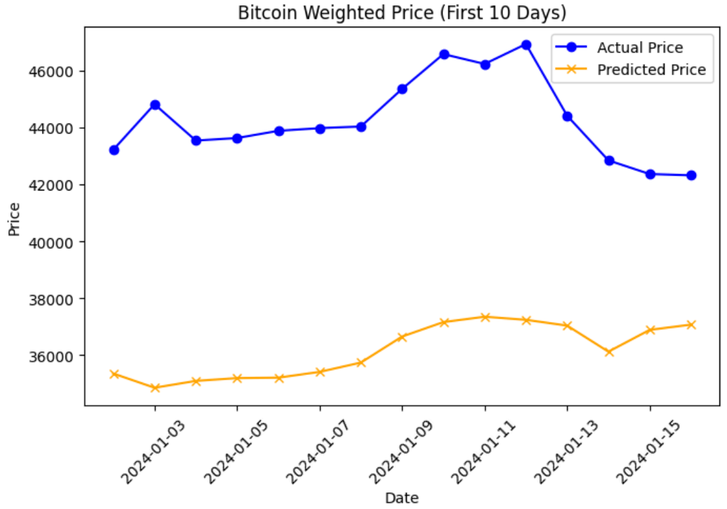


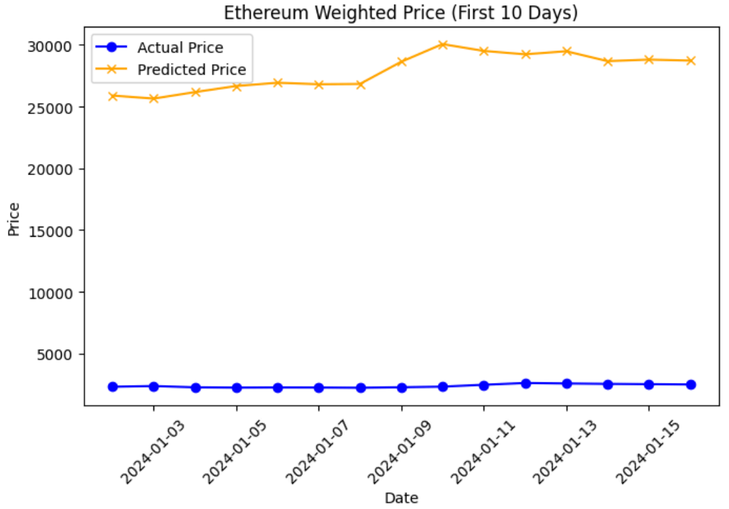
LSTM Model is the best-fit model

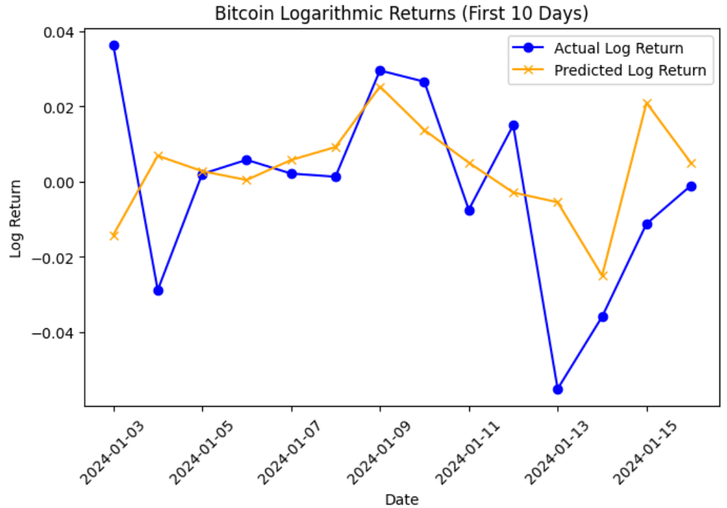
Test RMSE for LSTM on ETH: **39.904**

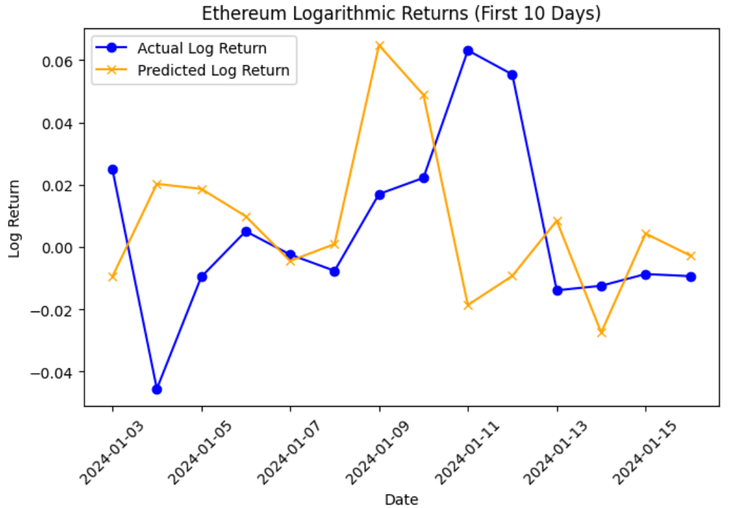
COMPARISON - BTC/ETH

- LSTM Model was used for predicting future prices of Bitcoin and Ethereum.
- Logarithmic Returns were plotted against the actual logarithmic returns.









CONCLUSION

- The LSTM model’s predictions exhibited the lowest RMSE, highlighting its superiority over traditional models
- Ethereum is identified as the more profitable investment choice, offering higher returns compared to Bitcoin over the observed period.

FUTURE WORK

- Inclusion of Additional Cryptocurrencies.
- Exploring Hybrid Models.
- Cross-Asset Analysis

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THANK YOU